

PROJ406 - Capstone Project - Deliverable 4

Group	2
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Solution Type	Power BI analytical report supported by Python data preparation and modeling

This project builds a practical analytical solution to evaluate which Alberta wells tend to remain productive longer, which patterns are linked to stronger performance, and which wells show early warning signs of decline or integrity risk. The solution combines well attributes, production history, and integrity-related data into a Power BI model that supports descriptive analysis, risk screening, and business-facing decisions.

1. Project Overview

This capstone project's goal is to tackle a significant business issue facing the oil and gas sector: determining which wells are most likely to continue producing in the long run and which might be vulnerable to early decline or operational failure. Organizations must have a trustworthy and data-driven method of assessing long-term asset performance since drilling decisions in capital-intensive sectors like oil and gas entail a large financial commitment.

With the help of Python-based data preparation routines, this project creates an analytical solution that integrates integrity indicators, well characteristics, and historical production measurements into a single Power BI display. The goal is to turn disjointed raw datasets into an analytical result that can be used for capital planning, risk screening, and production forecasting.

The absence of an integrated framework that enables decision-makers to differentiate between long-lasting wells, average-performing wells, and wells exhibiting early operational risk is the business issue being addressed. This issue is significant because incorrect lifespan estimations can result in inefficient drilling techniques, poor capital allocation, and higher maintenance or abandonment expenses.

2. Objectives / Business Questions

The objectives of this project are directly aligned with the proposal and focus on generating actionable insights that support operational and strategic decision-making. The analysis seeks to identify the well attributes most closely linked to long-term success, estimate average productive duration across different well types and regions, and detect early warning signs of production decline or operational issues.

The key business questions guiding this project are as follows:

1. What well attributes are most strongly associated with historically successful and long-lasting wells?
2. What is the average productive duration of oil and gas wells across different well types and geographic areas?
3. Which wells show early signs of decline or integrity-related risks?
4. Can wells be classified into long-lasting, average-performing, and early-failure groups using early production data?
5. Finally, how can production and integrity indicators be combined to strengthen long-term forecasting accuracy?

The final solution addresses each of these objectives through interactive visuals, DAX measures, and risk segmentation logic built into the Power BI report.

3. Data Understanding

The analysis is based on a blended dataset created from multiple operational and production data sources. The primary grain of analysis is the individual well, while monthly production records provide the time-series foundation required to calculate lifespan and production trends.

Key Source Table

Table / Source	Primary grain	Purpose in model	Example key
ST37 Well List	One row per well	Master well attributes, classification fields, identifiers, and location context	UWI / Licence
Petrinex master fact	Well-month	Production history used for duration, trend, and performance measures	Well + month
Well casing failure data	Well or incident	Integrity risk support and operational warning context	Licence / UWI

Support flags / auxiliary tables	Varies	Lookup fields, grouping logic, and optional enrichment for analysis	Mapped IDs
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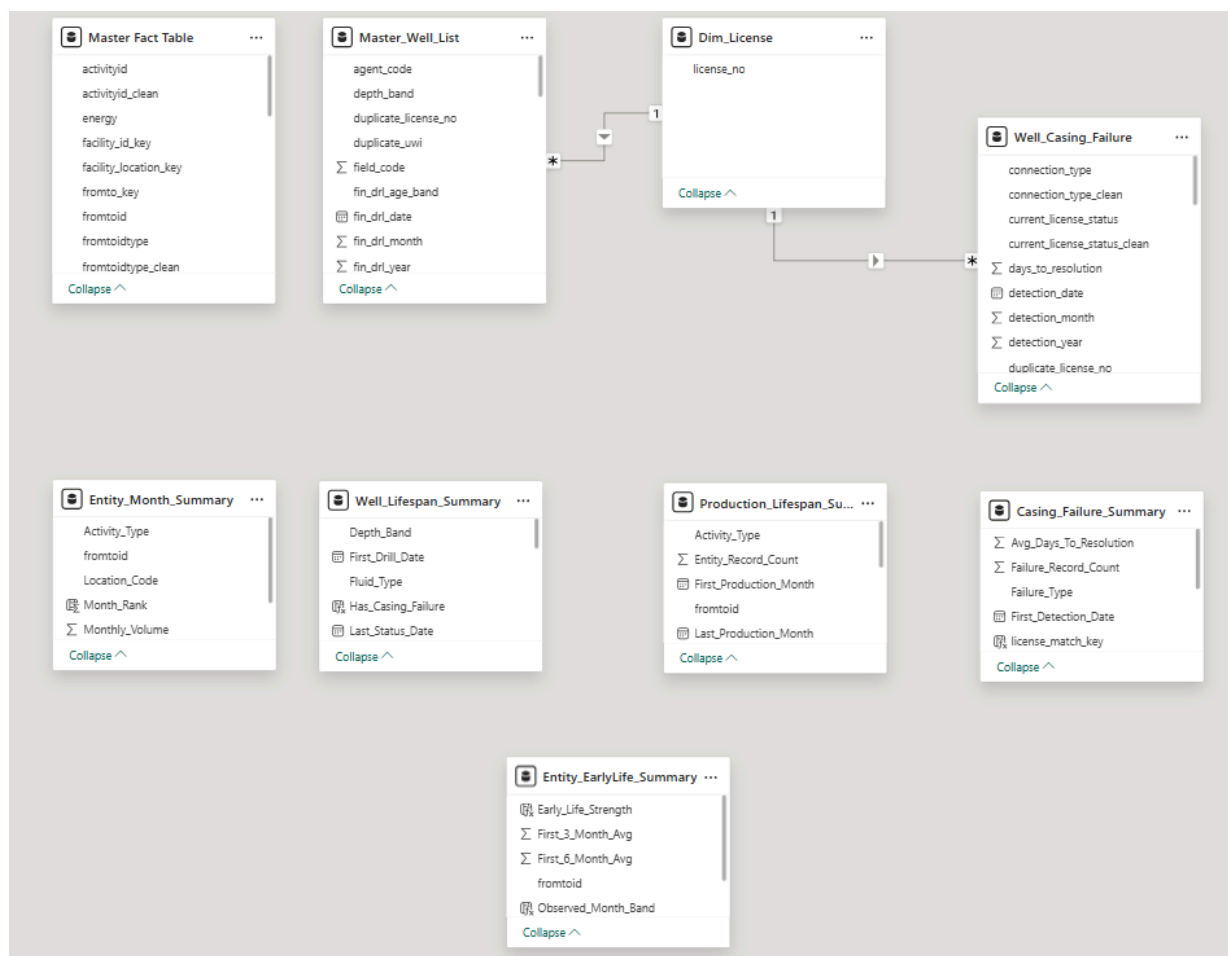
Data Dictionary

Field	Data type	Table	Description
UWI / Well ID	Text	Well List / Fact	Unique well identifier used for joins and tracking.
Licence No	Text/Whole	Well List	Regulatory or operational identifier used in source matching.
Well Type	Text	Well List	Categorizes wells for lifespan and production comparison.
Field / Area	Text	Well List	Geographic grouping used for regional analysis.
Spud / Onstream date	Date	Well List	Supports age and timeline calculations where available.
Production Month	Date	Petrinex	Month-level timestamp for production facts.
Oil Volume	Decimal	Petrinex	Monthly oil output used for productivity analysis.
Gas Volume	Decimal	Petrinex	Monthly gas output used for productivity analysis.
Water / Other volumes	Decimal	Petrinex	Used where relevant for supporting operational context.
Productive Duration	Calculated	Model measure	Calculated lifespan based on observed production activity.
Early Decline Flag	Calculated	Model measure / class	Identifies wells showing weak early pattern or drop-off.
Integrity Risk Flag	Binary/Text	Casing failure / support	Indicates known integrity event or elevated risk context.

4. Data Model Relationships

In order to facilitate effective filtering and analysis, the final Power BI data model has a useful star-like topology. The Petrinex table is the fact table that houses monthly production statistics, while the Well List table is the main descriptive dimension table that contains the master attributes for every well. The model is enhanced with failure and risk-related context by connecting integrity and operational support tables using the same well-level IDs.

To guarantee consistency across source systems, relationships were mostly constructed using the UWI/Well ID field. Without repeating descriptive variables, this model design enables filtering from descriptive attributes like well type and geographic area into the lower-grain monthly production data. To enhance report usability, the structure was purposefully kept tidy and business-focused.



5. Solution / Analytical Product

An interactive Power BI dashboard designed to facilitate lifecycle analysis, performance comparison, and early-risk identification is the finished analytical product. The dashboard was designed to give decision-makers an easy-to-use method of analyzing vast amounts of operational and production data.

The initial step in the process is to use Power Query and Python transforms to clean and standardize the raw source files. The master well list, production facts, and integrity support tables are linked using well-level identifiers once they have been standardized. After that, DAX measures were

developed to compute risk-related classifications, cumulative production indicators, and productive duration.

Users can assess performance by type, area, and risk category using the dashboard's comparative graphics. Additionally, it features category-based breakdowns that divide wells into long-lasting, early-decline screening perspectives.



6. Analytical Process

The analytical workflow for this project was designed as a structured, multi-stage pipeline, progressing from raw data preparation to actionable business insights.

Stage 1: Data Cleaning and Validation

To guarantee the dataset's consistency and dependability, the procedure started with a thorough data cleansing in Power Query. This involved fixing data types (including dates and numeric production values), unifying column names and formats across several source tables, and verifying important identifiers like FromToID and Well License. Logic-based flags, such as `potential_duplicate_fact_row`, were used to identify duplicate fact records and evaluate them prior to removal.

Incomplete and missing records were also handled in a methodical manner. Only analytically significant data—that is, rows with `has_any_measure = Yes`, `has_missing_important_data = No`, and `potential_duplicate_fact_row = False`—were retained using filters. This made sure the model was only loaded with high-quality, analysis-ready data.

Stage 2: Data Transformation, Integration, and Feature Engineering

Several datasets, including production data and well metadata, were combined throughout the transformation stage to produce a single analytical model. Cross-table analysis between well features and production performance was made possible by the establishment of relationships utilizing consistent keys like FromToID and Well License.

The dataset was improved in large part thanks to feature engineering. To obtain useful data like these, new computed columns and DAX measures were developed.

- Productive duration of wells
- Production decline indicators
- Aggregated production volumes over time
- Well classification attributes

In order to facilitate early warning analysis, custom risk flags were also created. These flags laid the groundwork for predictive-style classification by identifying wells with characteristics like deteriorating production trends, inconsistent output, or missing operational signals.

Stage 3: Analytical Modeling and Insight Generation

Extracting insights in line with the capstone's business questions was the main goal of the last phase. This comprised:

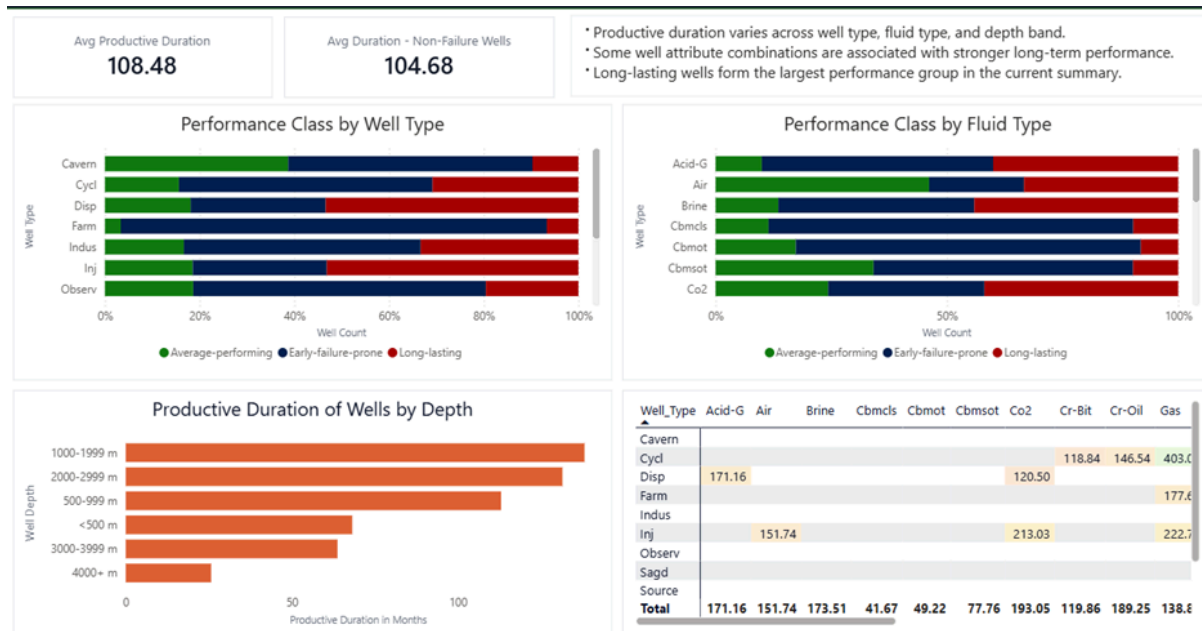
- Descriptive Analysis: Summarizing total well counts, production volumes, and average productive durations to establish baseline performance.
- Trend Analysis: Evaluating production behavior over time to identify decline curves and lifecycle patterns of wells.
- Comparative Analysis: Segmenting wells by type and geographic regions to uncover performance differences and operational trends.
- Classification and Risk Screening: Applying rule-based logic to categorize wells into groups such as long-lasting, average-performing, or high-risk for early failure.

The study transitioned from static reporting to decision-support analytics by merging production data with well integrity indications. Stakeholders can use the final model to find high-performing qualities, spot operational risk early on, and assist with data-driven long-term forecasting.

Business Questions and Answers:

1. Well Performance Overview

What well attributes are most strongly associated with historically successful and long-lasting wells?



This dashboard analyzes how well type, fluid type, and depth influence well performance classification.

Performance Class by Well Type

This chart compares the distribution of long-lasting, average, and early-failure wells across different well types.

It shows that certain well types have a higher proportion of long-lasting wells, while others are more prone to early failure, indicating that well design and operational purpose directly impact performance.

Performance Class by Fluid Type

This visual highlights variation in performance across different fluid types.

Some fluid types demonstrate stronger long-term outcomes, while others show higher early-failure rates, suggesting that fluid composition plays a significant role in production sustainability.

Productive Duration of Wells by Depth

This chart presents the average productive lifespan across depth ranges.

Wells in the mid-depth range (1000–3000 m) exhibit the highest durations, while very shallow and deep wells show reduced performance, indicating that depth optimization is critical for maximizing well lifespan.

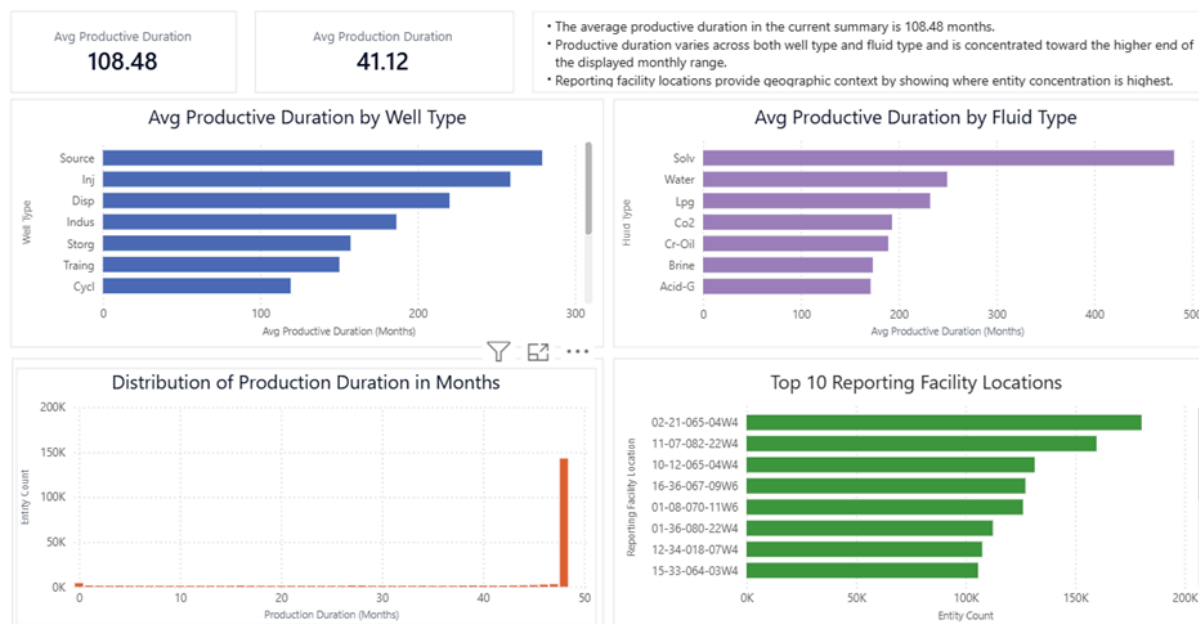
These metrics confirm that long-lasting wells contribute significantly to overall performance averages.

Key Takeaway

Well-success is influenced by a combination of attributes, particularly well type, fluid type, and depth, which together determine long-term performance outcomes.

2. Production Duration

What is the average productive duration of oil and gas wells across different well types and geographic areas?



This dashboard evaluates the average productive lifespan of wells and how it varies across well types, fluid types, and geographic locations.

Visual Analysis

Average Productive Duration KPI

The overall average productive duration is approximately 108 months, providing a baseline for comparison across categories.

Average Duration by Well Type

This chart shows significant variation across well types.

Source and Injection wells demonstrate longer lifespans, while cyclic wells have shorter durations, indicating that operational purpose impacts longevity.

Average Duration by Fluid Type

This visual highlights differences across fluid types.

Solvent-based wells show the highest average duration, followed by water and gas-based wells, suggesting that fluid handling and recovery methods influence sustained production.

Distribution of Production Duration

The histogram shows that most wells are concentrated toward the higher end of the duration range, indicating that a majority of wells maintain production for extended periods.

Top Reporting Facility Locations

This chart identifies regions with the highest well counts.

It provides geographic context, showing that production activity is concentrated in specific locations, which may influence overall averages and operational strategies.

Key Takeaway

Although the average lifespan is relatively high, well longevity varies significantly across well types, fluid types, and locations, emphasizing the importance of operational and geographic factors.

3. Early Decline and Risk Detection

Which wells show early signs of decline or integrity-related risks?



This dashboard identifies failure-prone wells and examines how early production performance can indicate future operational risk. The objective is to determine whether the initial production behaviour can be used as an indicator of long term well outcomes.

Visual Analysis

Failure vs Non-Failure Wells

This chart shows that failure wells represent a smaller portion (9K) compared to non-failure wells (345K).

Although fewer in number, these wells represent critical operational risks.

Early-Life Performance of Wells

This visual categorizes wells into strong and slower starts based on early production levels.

A larger number of wells fall under the “Slower Start” category (111,000 wells) compared to those with a “Strong Start” (92,000 wells). This suggests that a substantial portion of wells begin with lower production levels, which may indicate higher risk.

Productive Duration: Failure vs Non-Failure Wells

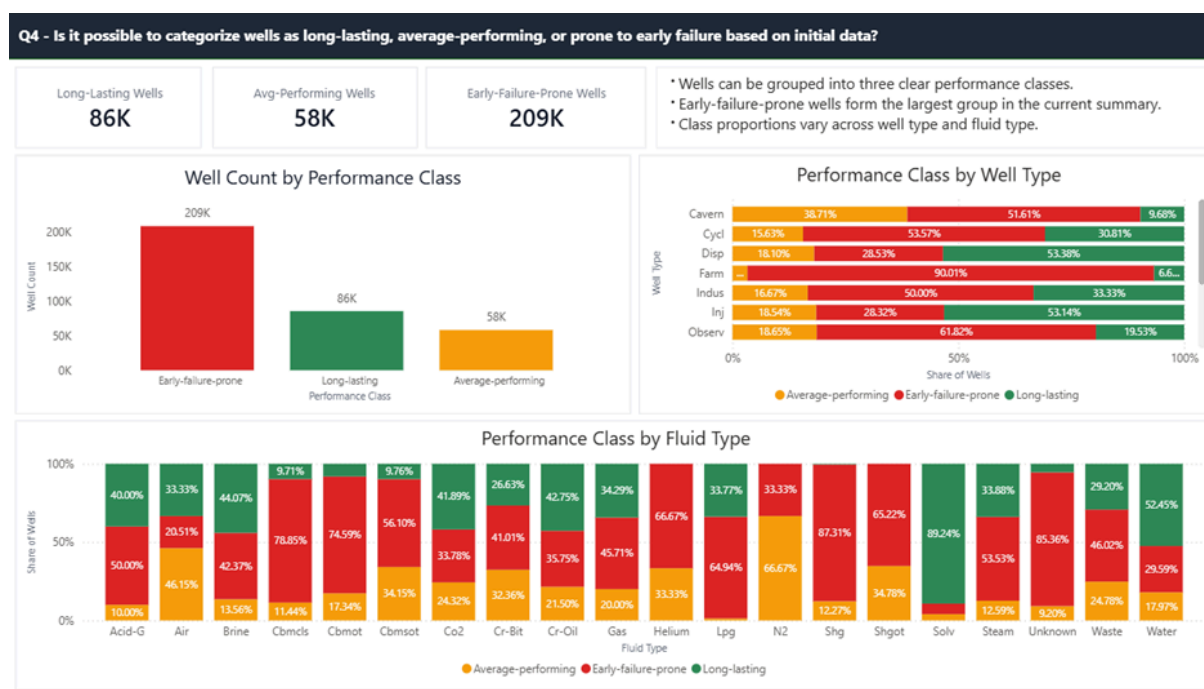
The relationship between early performance and long-term outcomes is clearly demonstrated in this chart. Failure wells have an average productive duration of approximately 105 months, whereas non-failure wells average around 255 months. This significant difference of over 140 months confirms that wells with early production challenges tend to have considerably shorter lifespans.

Key Takeaway

Early production performance is a strong predictor of well outcomes. Wells that exhibit a slower production start and lower initial output are more likely to experience early decline and operational issues, making early-life production a critical indicator for identifying high-risk wells.

4. Well Classification

Can wells be classified into long-lasting, average-performing, and early-failure groups using early production data?



This dashboard evaluates whether wells can be reliably classified into performance categories long-lasting, average-performing, and early-failure-prone—using early-stage production and operational data. The goal is to determine if initial indicators can support predictive segmentation.

Well Count by Performance Class

This chart shows the distribution of wells across three categories:

- Early-failure-prone (209K)
- Long-lasting (86K)
- Average-performing (58K)

It highlights that early-failure-prone wells form the largest group.

Performance Class by Well Type

This visual shows how classification varies across well types.

Some well types have a higher share of long-lasting wells, while others are dominated by early failures, indicating that well type influences classification outcomes.

Performance Class by Fluid Type

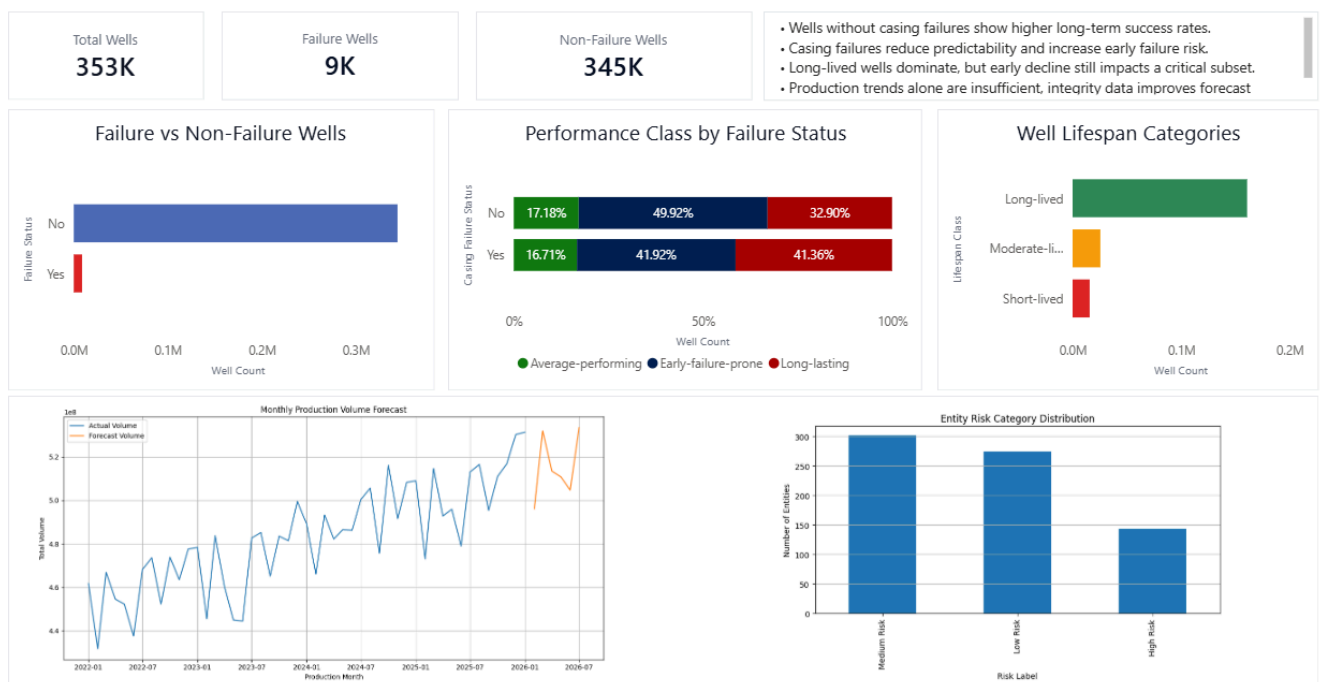
This chart demonstrates variation in performance categories across fluid types.

Certain fluid types are more associated with long-lasting wells, while others show higher early-failure proportions, reinforcing the role of fluid characteristics in performance segmentation.

Key Takeaway

The results demonstrate that wells can be reliably categorized into long-lasting, average-performing, and early-failure-prone groups using initial data. This classification is primarily driven by combining early production performance, decline rates, productive duration, well type, and fluid type into a unified classification framework. Instead of using a single metric, a pattern-based approach was applied to evaluate how these factors interact. Wells with stable production and low decline were classified as long-lasting, while those with high early decline and short lifespan were categorized as early-failure-prone. Moderate patterns were grouped as average-performing. This multi-factor approach ensured a reliable and realistic segmentation of well performance based on initial data.

5. Finally, how can production and integrity indicators be combined to strengthen long-term forecasting accuracy?



This dashboard evaluates how integrating production metrics with well integrity indicators improves the ability to assess and forecast long-term well efficacy.

The objective is to move beyond single-metric evaluation and determine whether a combined view of performance and operational health provides a more reliable foundation for decision-making.

Integrated Performance View (Production + Integrity Signals)

This view combines key production measures such as cumulative output, decline trends, and productive duration with operational integrity indicators.

It highlights that wells with strong production alone do not always translate to long-term success—those with stable production and consistent integrity signals show significantly better lifecycle performance.

Well Performance by Integrity Risk Levels

This visualization segments wells based on risk flags derived from early indicators.

Wells categorized as low-risk demonstrate more stable production trends and longer productive durations, while high-risk wells show early decline patterns, irregular production behavior, or data inconsistencies—indicating potential operational issues.

Trend Analysis of Production vs Risk Signals

This chart compares production trajectories across different risk categories over time.

It shows that wells flagged early for risk tend to experience sharper decline curves and shorter lifespans, whereas low-risk wells maintain more consistent output, reinforcing the predictive value of combining these signals.

Key Takeaway

The analysis demonstrates that combining production metrics with well integrity indicators significantly enhances the ability to evaluate and forecast long-term well performance. Rather than relying solely on production volume, this approach incorporates decline behavior, data reliability, and operational risk signals into a unified framework.

Wells exhibiting stable production trends alongside low-risk integrity profiles are more likely to sustain long-term productivity. In contrast, wells with early warning signals—such as rapid decline, inconsistent output, or flagged data issues—are more prone to underperformance.

This integrated, multi-factor approach provides a more accurate and realistic assessment of well efficacy, enabling better forecasting, early risk detection, and more informed operational and investment decisions.

7. Findings / Results

The final solution effectively addresses the core business questions by transforming fragmented and disconnected operational datasets into a unified, insight-driven analytical model. By integrating production data with well-level attributes, the analysis provides a comprehensive view of well performance across its lifecycle.

The model identifies key well characteristics that are consistently associated with long-term productivity, including patterns observed across well types, completion styles, and geographic regions. It also quantifies differences in productive duration, highlighting how certain categories of wells sustain output longer than others, while some exhibit faster decline cycles.

A major value-add of the solution is its ability to proactively detect risk. Through trend analysis and rule-based screening, the dashboard flags wells that show early signs of production decline, irregular output patterns, or potential operational integrity concerns. These early warning indicators allow stakeholders to shift from reactive monitoring to proactive decision-making.

Additionally, by combining production metrics with well integrity signals, the solution moves beyond traditional volume-based evaluation. This integrated approach enables a more accurate assessment of

long-term well efficacy, supporting better forecasting, improved asset management, and more informed investment decisions.

Overall, the dashboard is not just a reporting tool—it functions as a decision-support system that helps identify high-performing well profiles, isolate underperforming or at-risk assets, and guide strategic planning for future drilling and operational optimization.

8. Business Impact / Recommendations

The business value of this solution lies in its ability to support evidence-based operational and investment decisions. By identifying high-performing well categories and early-risk cases, the dashboard can support better capital allocation, production planning, and risk mitigation strategies.

It is recommended that organizations prioritize drilling strategies based on historically stable well profiles, closely monitor wells flagged as early-risk, and continue integrating integrity data into forecasting models. The solution can also serve as a foundation for future predictive modeling and operational scorecards.

9. Changes Made from Deliverable 1

Compared to Deliverable 1, the scope of the project was refined to focus on a stronger business-facing Power BI solution. Advanced predictive modeling components were repositioned as a future extension, while the dashboard became the primary analytical product for decision support and presentation readiness.

10. Risks / Rewards

One of the main risks in this project was the size and inconsistency of the source data, particularly when integrating multiple operational datasets. Additional limitations included time constraints and the availability of complete integrity-related variables. One insights we took really quickly while working on this project was, we got to know that real industrial data are really messy.

The reward, however, is a scalable and practical analytical framework that improves forecasting confidence, strengthens risk identification, and supports strategic production decisions.

11. Conclusion

In conclusion, this project successfully developed a decision-support dashboard that enables oil and gas organizations to better understand well lifespan, production stability, and early operational risk. By combining production history with integrity indicators, the solution provides actionable insights that support stronger operational planning, investment decisions, and future analytical extensions.

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